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CASE REPORT

PATHOLOGY/BIOLOGY

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Sudden Death Due to Primary Intraventricular Hemorrhage: Report of Two Cases

ABSTRACT: Primary intraventricular hemorrhage (PIVH) is a rare type of stroke defined as bleeding within the ventricles of the brain without any associated parenchymal hemorrhage. Here, we reported two cases of sudden death due to PIVH. One of the patients was found dead under a highway bridge without witnesses, and the other patient was hospitalized with hemorrhage in the ventricular system, as revealed by a head computed tomography scan. In these two patients, autopsy and macroscopic examination only showed hemorrhages in the ventricular system without any traumatic brain injury or other intraparenchymal hemorrhage. The sources of bleeding for both patients were ultimately confirmed as ruptured brain arteriovenous malformations located in the subventricular zone. We reported these cases to broaden our understanding of sudden death associated with PIVH, especially when caused by brain arteriovenous malformation. We also summarized the essential details of the diagnoses and available technical methods for PIVH cases.

KEYWORDS: forensic science, forensic pathology, primary intraventricular hemorrhage, brain arteriovenous malformation, sudden death, immunohistochemistry, Elastic-Van Gieson stain

Intraventricular hemorrhage (IVH) is customarily classified into primary intraventricular hemorrhage (PIVH) and secondary intraventricular hemorrhage (SIVH), depending on the sources of bleeding (1). PIVH is defined as ventricle hemorrhage caused by ruptures of the choroid plexus or subventricular zone (SVZ), while SIVH is frequently the result of intracerebral or subarachnoid hemorrhage (SAH) (1,2). PIVH and SIVH are present in approximately 30% and 70% of the IVHs, respectively (3,4). PIVH accounted for only 3% of the intracranial hemorrhage (ICH) (5). A study exploring the features of PIVH in a Chinese population showed that 67 patients were diagnosed with PIVH, which constituted 3.2% of the hemorrhagic stroke patients in the study (6). The pathogenesis of PIVH has been associated with hypertension, arteriovenous malformation (AVM), microaneurysm, Moyamoya disease, arteriovenous fistula, congenital cerebral edema, choroid plexus tumor, and coagulation mechanism disturbances (1,7). AVM and microaneurysms were two of the most common contributing factors (7).

Here, we present two cases of death due to PIVH induced by ruptured brain arteriovenous malformation in the SVZ.

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Case Report

Case 1

A 26-year-old man was found dead under a highway bridge without witnesses.

At autopsy, there were skin abrasions on the right forehead and dorsum nasi. The brain weighed 1400 g. There were diffuse SAHs on the brain surface, and cerebellar tonsillar herniation was present. Hemorrhages were seen around the brainstem. Ventricular systems were dilated by hemorrhage, and the hemorrhage in the right lateral ventricle was more serious.

Upon histological examination, cerebral edema was notable, and there were extensive perivascular hemorrhages under the superficial brain tissue of SVZ with hemosiderin deposited. Neurons around the hemorrhage foci were necrotic, and the astrocytes were enlarged and became vesicular with the cytoplasm stained bright pink. Nidus in SVZ at the bottom of the right lateral ventricle was composed of abnormal vascular plexus as intertwined caliber hyaline thick-walled vessels, with focal hemorrhages. The continuation of the blood vessel walls with different thicknesses could be seen in some blood vessels. Elastic-Van Gieson (EVG) staining showed a thick internal elastic lamina layer in large arteries and dysplastic veins with less staining for elastic fibers in AVM nidus (Fig. 1a). Immunohistochemical staining for alpha-smooth muscle actin (α -SMA) demonstrated that α -SMA was expressed in the smooth muscle cells of the AVM nidus, indicating immature molecular construction of vessels (Fig. 1b).

Case 2

A 39-year-old man was admitted to the hospital for unexpected headache and loss of consciousness. The patient was in

deep coma, and the bilateral pupil diameter was 1.5 mm. His blood pressure was 200/110 mmHg. A cranial CT scan showed hemorrhages in the ventricular system. The man was pronounced dead after hospitalization.

At autopsy, the brain weighed 1512 g with severe swelling. Transtentorial and cerebellar tonsillar herniation was present. A massive hemorrhage and blood clot had filled the bilateral, lateral, third, and fourth ventricles. Splinter hemorrhage and blood clot in the left stria terminalis and caudate nucleus were also notable.

On microscopic examination, congestion of the subarachnoid blood vessels was characteristic. Neurons appeared necrotic, and neuronophagia was present. There were multiple infarctions and hemorrhages in the basal ganglia. Around the hemorrhagic region of the left stria terminalis and caudate nucleus, a small vascular plexus with uneven thickness of the walls was observed. Hemosiderin was deposited in the tissue surrounding vessels, combined with hemorrhages and necrosis. Hemorrhages accompanied by reactive astrogliosis were clear in the SVZ of the lateral ventricle. EVG staining and α -SMA staining showed similar results compared with the results of case 1 (Fig. 2a,b).

Discussion

Primary intraventricular hemorrhage was described as bleeding in the ventricular system without a discernable parenchymal

component, which was first reported by Sanders in 1881 (5). Based on the clinical manifestation, PIVH can be subdivided into two types as follows: type 1, sudden coma or unconsciousness, nausea, vomiting, nuchal rigidity, severe headache, and lethal brainstem dysfunction; type 2, fluctuant or progressive symptoms, including increased intracranial pressure accompanied by headache, vomiting, conscious disturbance, somnolence, and then loss of consciousness after hours to a few days (5,8,9). The former type is generally perceived as a typical PIVH. Before the maturity of clinical imaging technology, such as CT, magnetic resonance imaging (MRI), and angiography, the clinical diagnosis of PIVH was difficult; hence, histopathologic examination at autopsy was often required (7).

In case 2, the patient was admitted to the hospital for sudden headache and unconsciousness. A cranial CT scan showed intraventricular hemorrhages but no ICH. PIVH was established as the exclusionary diagnosis after autopsy ruled out traumatic brain injury. However, in case 1, diagnosis was essentially confirmed in accordance with an autopsy, as no antemortem symptoms were obtained due to the absence of witnesses or medical records. External examination showed only three abrasions on the head and face. Moreover, there were no scalp and skull injuries, cerebral contusions, or lacerations. Multifocal SAHs were observed on the surface of the cerebrum and brainstem, and a massive hemorrhage was observed in each ventricle. Since the ventricular system and subarachnoid space are connected through

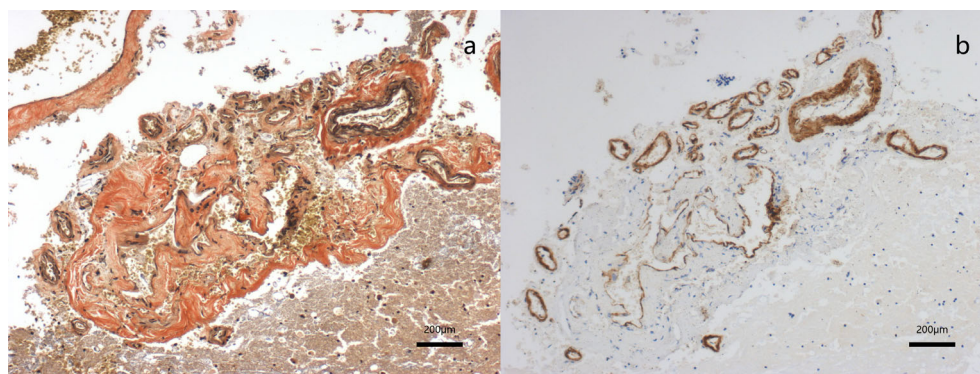


FIG. 1—(a) Vascular plexus is formed by blood vessels with different diameters, some blood vessels are obviously deformed, and parts of vascular wall are uneven and mainly composed of collagen fibers (Elastic-Van Gieson stain, $\times 100$). (b) The thickness of the abnormal vessel wall is uneven, and the smooth muscle layer is significantly thin and discontinuous (α -smooth muscle actin stain, $\times 100$). [Color figure can be viewed at wileyonlinelibrary.com]

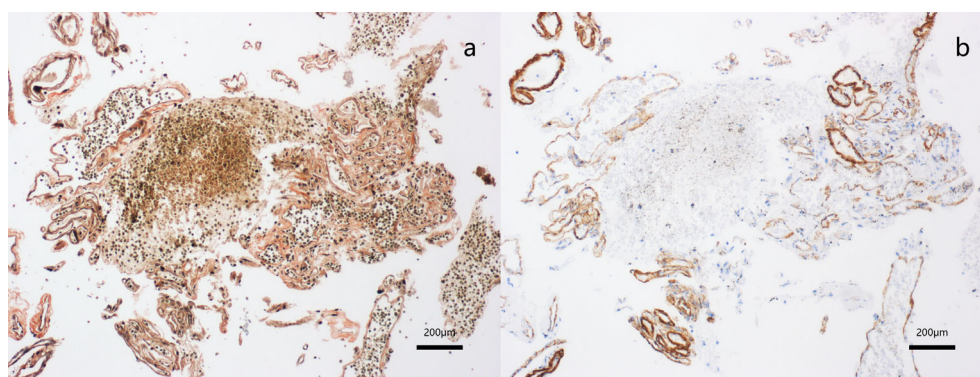


FIG. 2—(a) Intertwined deformed vascular mass can be observed in the hemorrhage foci (Elastic-Van Gieson stain, $\times 100$). (b) The vessel in the lesion is different in size, and the continuation of the arterial wall and the venous wall can be observed in some vessels (α -smooth muscle actin, $\times 100$). [Color figure can be viewed at wileyonlinelibrary.com]

median aperture of the fourth ventricle, approximately 45% of spontaneous ICHs and 25% of aneurysmal SAHs extend into the ventricles (10–12). PIVH used to be misdiagnosed as SAH due to the similarity in symptoms and in CSF examination until the popularity of CT images (13). Therefore, clarifying the relationship between IVH and SAH is a key issue that needs to be resolved urgently.

Subarachnoid hemorrhage may occur spontaneously or as a result of head trauma, with the former also regarded as pathologic SAH. Traumatic SAH usually occurs near the site of a skull fracture or intracerebral contusion. The common etiological factors of spontaneous SAH include cerebrovascular diseases, such as aneurysms and AVM, which overlap with PIVH (1,14). Thus, the location of the bleeding site is the crux of differential diagnosis for spontaneous SAH and PIVH. In patient 1, traumatic SAH was excluded through the autopsy. Pathological examinations did not show a ruptured aneurysm or vascular malformation in the subarachnoid space. Focal hemorrhages originated from brain tissues around the intertwining vascular plexus in the right lateral ventricular wall. Therefore, it was logical to conclude the abnormal vascular plexus in the SVZ of the right lateral ventricle as the source of bleeding, which met the diagnostic criteria of PIVH and indicated that the SAH was secondary.

In these two cases, bleeding sites were both located in the SVZ of the lateral ventricle, and abnormal vascular plexuses with uneven thickness of the walls were found in the hemorrhages. The abovementioned characteristics matched the pathologic features of an AVM rupture. An AVM can be divided into small-sized (<3.0 cm), middle-sized (3.0–6.0 cm), and large-sized (>6.0 cm) lesions (15). The lesion size and location may affect the possibility of a secondary rupture of an AVM. Larger-sized AVMs tend to trigger epilepsy by damaging surrounding tissues, while smaller-sized AVMs usually cause hemorrhages (16). Deep brain location is an independent predictor of a subsequent hemorrhage (17). Moreover, it was demonstrated that the hemorrhagic tendency of periventricular and intraventricular AVMs was relatively high (18). Indeed, these features were observed in the two cases we presented here. In addition, since the compositions of elastic fibers and smooth muscles in the vessel wall are the main indicators for identifying arteries and veins, EVG staining and α -SMA staining could help diagnose AVM. Finally, the cause of death for patient 1 was ascribed to PIVH that was induced by the ruptured AVM in the SVZ, and the mild skin abrasions might be the result of falling down due to unconsciousness. For patient 2, PIVH was more likely to be incurred by ruptured AVM of stria terminalis and caudate nucleus.

Conclusion

Primary intraventricular hemorrhage should be considered only when traumatic ventricular hemorrhage is excluded. Hemorrhage caused by choroid plexus hemorrhage, intraventricular

tumor, or tissue bleeding of the SVZ <1.5 cm in depth must be ascertained. Obtaining proof of an SAH combined with ventricular hemorrhage in an autopsy should be performed before formalin fixation. The presence of AVM, microaneurysm, or tumor beneath the ventricular walls should also be taken into account. The morphology and position of the most serious ventricle hemorrhage should be examined with serial sections to avoid misjudgment. Medical reports, such as CT or MRI records, are important to acquire. The use of anticoagulants and alcohol intake should also be considered. EVG staining and α -SMA staining can help identify AVM lesions.

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